The AI-Assisted Software Development Life Cycle (SDLC)

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This document outlines the standard engineering practices, CI/CD requirements, and AI-assisted workflows required for all project implementations. It covers the maintenance of Agent Skills, a 'Second Brain' knowledge base, GitHub team workflows, and the phased deliverable stages common to both the YouTube MCP Server and the Financial Ledger API projects.

1. Second Brain Requirements

- **Structure (PARA):** Organise notes by actionability. brain/inbox/ (unsorted), brain/projects/ (active deadlines), brain/areas/ (ongoing responsibilities), brain/resources/ (reference material), brain/archive/ (inactive). brain/index.md is the entry point. brain/log.md records all runs.
- **Workflow Skills:** Maintain the brain using three skills: brain-ingest (add new material/summaries), brain-refine (move to PARA, merge duplicates, fix links), and brain-query (answer questions, draft documents like SRS and ADRs).
- **SB-1:** Link notes to each other with [[wikilinks]], and to requirement IDs, issues, and PRs.
- **SB-2:** Write a short review note at the end of each stage: what worked, what did not, and what to change.
- **SB-3:** Every ADR and every skill shall cite at least one distilled note.
- **SB-4:** Keep an AGENTS.md (or equivalent) at the repository root.
- **SB-5:** Changes to the brain, including AI changes, go through pull requests.
- **SB-6:** Every team member shall author notes. Contribution must be visible in commit history.

2. Software Engineering Requirements

- **SE-1:** Produce and maintain an SRS containing functional (FR), non-functional (NFR), and skill requirements (SK).
- **SE-2:** Produce system architecture/design docs with appropriate UML/ER diagrams.
- **SE-3:** Record significant decisions as Architecture Decision Records (ADRs) in docs/adr/, citing brain notes.
- **SE-4:** Produce a development plan and task breakdown.
- **SE-5:** Keep a traceability matrix: requirement -> issue -> PR -> test -> skill -> brain note.
- **SE-6:** Isolate external dependencies behind a provider interface.

- **SE-7:** Produce user documentation and developer API documentation (e.g., OpenAPI).

3. Common Agent Skills

- **local-check:** Confirm that the project works on a team member's machine: 1. Build the project. 2. Run the automated test suite. 3. Start the local server. 4. Invoke one read-only fetch endpoint. 5. Report pass or fail for each step.

4. GitHub and Team Development

- **Version Control:** Meaningful incremental commits from every member. Documented branching strategy. GitHub Issues for all work/defects. Pull requests with at least one review for every change (code, skills, brain). Release tags and changelog.
- **CI Pipeline:** GitHub Actions (or equivalent) shall run on every PR: Build, lint, automated test suite, secret scanning, validation of every SKILL.md (frontmatter/naming), validation of brain notes (schema/broken links).

5. Common Testing and Quality Requirements

- **TQ-7:** Skill evaluations shall run before each stage submission. Record results in the test documentation.
- **TQ-8:** Maintain and update tests as the system changes. Each significant defect gets a regression test.

6. AI-Assisted Development Policy

- **Responsibility:** AI coding assistants are expected, but the team remains responsible for every line merged. A human shall review every AI-generated change in a PR.
- **Comprehension:** Every member shall be able to explain any merged code, skill, or document during project reviews.
- **Documentation:** Record significant AI sessions in the brain as ai-session notes (goal, what worked/failed).
- **Security:** Do not paste secrets, tokens, or credentials into any AI tool.

7. Expected Deliverables by Stage

- **Stage 1 - Requirements:** SRS (FR, NFR, SK). Brain set up as an Obsidian vault (PARA, index, log). brain-ingest skill used to add at least 10 research notes. AGENTS.md. Repository and CI skeleton.
- **Stage 2 - Design:** Architecture and design documentation. Diagrams. brain-refine and brain-query skills. First version of the local-check skill.

- **Stage 3 - Testing and CI:** Continuous testing and CI plan/results. The product-specific skill (video-to-playlist/simulate-transfer) with test results. Defect log with regression tests. Review note for each stage.
- **Stage 4 - Review:** Live demonstration of FRs in client. Run product skill end-to-end. Run local-check live. Run brain-query live. Walk through a requirement from brain note to ADR, code, and test. Final release tag.

8. Project Expectation

The objective is to show how a software engineering team takes a real-world problem and builds a maintainable product with sound requirements, design, implementation, testing, version control, collaboration, and documentation. Skills and the second brain serve to make the team's knowledge explicit and reusable.

9. Tools and References

- **Obsidian:** Editor for the brain/ vault: notes, links, and graph view.
- **How I Built My Second Brain with Obsidian + Claude Code:** How an AI agent ingests, structures, and queries an Obsidian vault through a schema file. Basis for the brain skills.
- **Agent Skills Format (agentskills.io):** Specification for SKILL.md folders.
- **Superpowers:** Skill set for coding assistants: brainstorming, planning, test-driven development, code review.
- **What is the Model Context Protocol?:** Introduction to MCP: how AI clients connect to servers that expose tools and data.
- **SQLite Documentation:** Reference for ACID transactions, transaction locking, and WAL mode.
- **Postman / Bruno / cURL:** Manual testing and verification of RESTful API endpoints.